

# IMMEDIATE SETTLEMENT OR ENDURING A STRIKE: THE CHOICE OF SIGNALS

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## ABSTRACT

Some labour contract negotiations involve strikes while most conclude with immediate settlement. This article offers a model of union-firm negotiation with private information to show that either strikes or immediate settlement will take place in the equilibrium. Different from most signalling literature where the signals are exogenously given, this article endogenizes the choice of signals. We compare two signals, the employment level and the strategic delay. We show that the low-revenue firm will choose the signal which gives it higher payoff while separating itself from the high-revenue firm.

*Keywords:* signaling, strategic delay, strike

*JEL classification numbers:* D82, J52, J64

## I. INTRODUCTION

An economic strike is a suspension of production when the firm and their employees negotiate and argue about the labour contract (Kennan, 2008). An important and stylized fact is that strikes do happen, although not often (Kennan and Wilson, 1990). Tracy (1986) and Kennan (1986) show that about 15 percent of contract renewals in the US manufacturing sector involve strikes. McConnell (1989) finds that this ratio varies from 10 to 40 percent depending on industries. Card (1990) indicates roughly 22 percent for Canadian data. Therefore, it is inadequate for models to obtain either only the result of unconditional strikes or only the outcome of immediate settlement. We need models capable of predicting both immediate settlement and strikes, with the immediate settlement in the majority of cases (Kennan and Wilson, 1989).

When does the negotiation end up with immediate settlement? When does a strike take place? We address these questions using a simple model of union-firm bargaining with private

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information. A labour union and a firm negotiate over the labour contract which specifies the wage and the employment level. The firm has private information about its revenue. When the low-revenue firm makes the offer, it can signal its type either by the employment level (and the corresponding wage), or by delay. More precisely, the low-revenue firm can distort the employment level downward enough (compared to the first-best) so that the high-revenue firm will not mimic; or alternatively, it can delay the offer long enough to separate itself from the high-revenue firm since delay is more costly for the latter. We show that based on the cost–benefit analysis from the low-revenue firm’s perspective, the employment level dominates delay as a signal under certain conditions; otherwise, delay dominates the employment level. On the one hand, when the firm uses the employment level as the credible signal, the union will accept it without delay, which means an immediate settlement. On the other hand, when the firm delays its offer to signal its type is costly, the outcome is a strike or enduring a strike (Kennan and Wilson, 1989; Kennan, 2008).

We then show some examples to illustrate our results. In the first example, where the revenue function is linear and the labour cost function is quadratic, we show the negotiation will end up with immediate settlement and no strike will happen. In the second example, where the revenue function remains linear while the labour cost becomes exponential, our numerical result indicates that immediate settlement with unemployment happens with large parameter values and strikes take place with only a small range of parameter values. We also extend our two-type model to the case of finite many types. We show that there exists a threshold type, and those types higher than the threshold type will signal by the employment level; those types lower than the threshold type will signal their types involving delay.

Delay has long been recognized as a signal in the bargaining literature (e.g., Admati and Perry, 1987). In the firm–union negotiation scenario, delay is interpreted as strikes or enduring strikes (Kennan and Wilson, 1989; Kennan, 2008). There is a large literature on the economics of labour market negotiation and strikes (see Kennan and Wilson, 1989, 1993 for surveys). Some of these papers predict immediate settlement, others posit unconditional strikes. Complementary to the literature, our model shows that both immediate settlement and strikes are possible in the equilibrium. Most of the existing literature focuses on one-dimensional negotiation, that is, the union and the firm bargaining over the wage. However, our model discusses multidimensional bargaining: the firm and the union negotiating over the wage and the employment level (Sen, 2000; Yao, 2012).

Cranton and Tracy (1992, 2003) develop a private information model of union contract negotiation including both screening and signalling elements. The uninformed union makes the first offer. If the offer is rejected, the union can choose between either strike or holdout. A firm which refuses the initial offer waits some time and then makes a counter-offer. Our model is different from theirs in the sense that the informed firm makes the first offer, and it chooses between immediate settlement or enduring a strike.

We present a signalling model here except that the signal is endogenously chosen. Starting from the pioneering work of Spence (1973), the idea of signalling has found prominent applications, for example, in labour economics, corporate finance, and industrial organization. In most literature, the signalling device is exogenously given. However, in many circumstances the principal actually has many ways to signal the same characteristic.<sup>1</sup> In our union–firm negotiation model, we endogenize the choice of signals. Our model nests two signals: the employment level (and the related wage) and the strategic delay. We show that the firm will choose the signal based on cost–benefit analysis. There are a few papers discussing the choice between signals.

<sup>1</sup> For example, the firm selling a high-quality good can signal the quality by advertising (Kihlstrom and Riordan, 1984), introductory pricing (Bagwell and Riordan, 1991), price plus advertising (Milgrom and Roberts, 1986), as well as warranties (Grossman, 1981).

Viswanathan (1995) presents a financial model which has both rights and outside equity issues as potential signals. He derives the optimal signalling schedule and characterizes the choices between rights and outside equity issues.

In Section II we lay out the model and show our results. Section III concludes.

## II. THE MODEL

A firm and a labour union negotiate over employment level  $L$  and total wage  $W$ . The union's labour cost of  $L$  is  $C(L)$ , which is a twice continuously differentiable function, and the payoff for the union is  $W - C(L)$ . The union has no private information, but the firm has private information about its own revenue, which could be either high or low and denoted by  $i \in \{l, h\}$ . The payoff for type  $i$  firm is  $\pi_i(W, L) = R_i(L) - W$ , where  $R(L)$  is twice continuously differentiable with  $R_h(L) > R_l(L)$  for all  $L$ , and  $R_h(L) - R_h(L') > R_l(L) - R_l(L')$  for any pair  $L$  and  $L'$ . Denote total surplus  $S_i = R_i(L) - C(L)$ , which is strictly concave and maximized by a finite value  $L_i^*$ , and let  $S_i^* \equiv S(L_i^*) > 0$  for  $i \in \{l, h\}$ . The union believes that the firm's revenue is high with probability  $\eta$ , and this prior belief is common knowledge.

In the negotiation game, the two parties make alternating offers with strategic delay (Rubinstein, 1982; Admati and Perry, 1987). Suppose the firm makes the first offer at time zero. Let  $\tau$  denote delay. The offer by the firm can be made either right away ( $\tau = 0$ ) or with a delay ( $\tau > 0$ ) after the negotiation begins. The union updates its belief about the firm's type on the basis of the offer and then responds. Let  $\rho \equiv \rho(W, L, \tau)$  be the union's posterior belief that the firm's revenue is high when the offer by the firm is  $(W, L, \tau)$ . If the union accepts the firm's offer, the game is over and both sides obtain whatever the contract entitles them. However, if the union rejects the offer, it makes a counter-offer with or without delay, which the firm decides to accept or reject. If the firm rejects the union's offer, it again proposes a contract, and so on and so forth until they reach an agreement. Let  $r$  be the discount rate, then the discount factor is  $\delta \equiv \exp(-r\Delta)$ , where  $\Delta$  is the minimal time that it takes for the relevant player to make a response to an existing offer and it is normalized to one.

If there is no private information, type  $i$  firm will make offer  $(W_i^*, L_i^*, 0)$ , where  $W_i^* = \frac{\delta}{1+\delta} S_i^* + C(L_i^*)$  and 0 means no delay, and the union will accept it without delay (Rubinstein, 1982). With incomplete information, we consider perfect Bayesian equilibria, which require that the strategies are optimal given the beliefs and the beliefs are updated using Bayes' rule whenever possible. We select plausible equilibria by imposing the 'intuitive criterion' (Cho and Kreps, 1987). In our two-type model, it is well-known that the intuitive criterion eliminates all equilibria but the least-cost separating equilibrium. Therefore, without loss of generality, our analysis focuses on this unique equilibrium.

### II.1. Analysis

We first discuss a special case: signalling for free. In this case, the first-best employment level is obtained and the negotiation is successful without delay.

*Lemma 1. In the negotiation game, when the following condition holds:*

$$S_h^* + \delta S_l^* \geq (1 + \delta)[R_h(L_l^*) - C(L_l^*)] \quad (\text{C1})$$

the high-type (low-type) will make offer  $(W_h^*, L_h^*, 0)$   $(W_l^*, L_l^*, 0)$ , and the union will accept it.<sup>2</sup>

As usual, the low-type here has no incentive to mimic the high-type. Under condition (C1), the high-type will not mimic the low-type either since the incentive compatibility constraints are satisfied by the first-best contracts  $(W_h^*, L_h^*, 0)$  and  $(W_l^*, L_l^*, 0)$ . Hence, each type of firm proposes its complete information contract, and the union accepts it. The intuition for this result is because the high-type faces a trade-off here. By faking to be the low-type, the high-type will pay less to the union. However, the total surplus also shrinks since less labour is hired. If the difference between the two type firms is sufficiently large, the high-type could obtain less by faking to be the low-type. In other words, (C1) is more easily satisfied when the difference between the two types is sufficiently large.

From now on we assume that (C1) is violated and

$$S_h^* + \delta S_l^* < (1 + \delta)[R_h(L_l^*) - C(L_l^*)] \quad (\text{C2})$$

Therefore, the high-type has incentive to mimic the low-type's complete information contract. It is in the low-type's interest to distinguish itself from the high-type by some costly signal.<sup>3</sup> As mentioned above, we focus on the least-cost separating equilibrium. We can easily obtain the following Lemma:

*Lemma 2. In the least-cost separating equilibrium, the high-type firm will make offer  $(W_h^*, L_h^*, 0)$ .*

The above Lemma is quite intuitive. Since the low-type has distinguished itself from the high-type by a costly signal in the least-cost separating equilibrium, there is no reason for the high-type to delay its offer or to distort the employment level. Otherwise, it would receive strictly less payoff.

As we mentioned above, the low-type can signal its type either by the employment level or by strategic delay. We first characterize the equilibrium when the low-type only signals by the employment level. Then we discuss the equilibrium which involves only delay. Finally, we compare the payoffs of the above two equilibria to determine which signal the firm will choose. We can implement this analysis strategy simply because the two signals are substitutes in our model.

When the low-type signals its type only by the employment level, it faces the following problem:

$$\begin{aligned} \max_{\{W_l, L_l\}} \quad & R_l(L_l) - W_l \\ \text{s.t.} \quad & W_l - C(L_l) \geq \frac{\delta}{1 + \delta} S_l^* \end{aligned} \quad (\text{PC}^1)$$

$$\frac{1}{1 + \delta} S_h^* \geq R_h(L_l) - W_l, \quad (\text{IC}^1)$$

$$R_l(L_l) - W_l > 0 \quad (\text{PC}^2)$$

<sup>2</sup> Condition (C1) is derived from the incentive compatibility constraint:  $\frac{1}{1 + \delta} S_h^* \geq R_h(L_l^*) - W_l^*$ , which guarantees that the high-type does not prefer the low-type firm's complete information contract.

<sup>3</sup> It can be shown that the pooling offer cannot exist in the equilibrium path (Sen, 2000; Yao, 2012).

where (PC<sup>1</sup>) is the participation constraint for the union, which guarantees that the union will take the offer. Since the low-type has revealed its private information, the union will demand the complete information continuation payoff, which is  $\frac{\delta}{1+\delta}S_l^*$ . (IC<sup>1</sup>) is the incentive constraint for the high-type, which ensures that the high-type will not mimic the low-type's offer. (PC<sup>2</sup>) makes sure that the low-type's payoff is positive after costly signalling. Notice that both (PC<sup>1</sup>) and (IC<sup>1</sup>) are binding at the optimum. Therefore, we can obtain the following result:

$$(W_l, L_l) = \begin{cases} (W'_l, L'_l), & \text{if } \pi_l(W'_l, L'_l) > 0 \\ (0, 0), & \text{otherwise,} \end{cases}$$

where  $L'_l$  is the smaller solution to  $R_h(L_l) - C(L_l) - \frac{1}{1+\delta}S_h^* - \frac{\delta}{1+\delta}S_l^* = 0$  and  $W'_l = C(L'_l) + \frac{\delta}{1+\delta}S_l^*$ , and  $\pi_l(W'_l, L'_l) = R_l(L'_l) - C(L'_l) - \frac{\delta}{1+\delta}S_l^* > 0$ . There is no delay involved of course. (0, 0) means that there is no offer from the firm simply because the firm cannot afford to signal by the employment level. It may delay the offer instead as we will discuss below. We thus obtain the following Proposition when  $\pi_l(W'_l, L'_l) > 0$ :

*Proposition 1. In the negotiation game, when the low-type firm signals by the employment level, there is a unique least-cost separating equilibrium: (1) the high-type firm always offers its complete-information contract  $(W_h^*, L_h^*, 0)$ ; (2) the low-type firm offers the contract  $(W'_l, L'_l, 0)$ ; (3) the union's beliefs are  $\rho(W_h^*, L_h^*, 0) = 1, \rho(W'_l, L'_l, 0) = 0$ ; (4) if a firm offers a contract  $(W'', L'', 0)$  such that  $\pi_h(W'', L'', 0) > \pi_h(W_h^*, L_h^*, 0)$  and  $\pi_l(W'', L'', 0) < \pi_l(W'_l, L'_l, 0)$ , then the out-of-equilibrium belief of the union is  $\rho'' = 1$ .*

When the low-type firm signals its type only by strategic delay, it faces the following program:

$$\begin{aligned} \max_{\{\tau\}} \quad & e^{-r\tau} \{R_l(L^*) - W_l^*\} \\ \text{s.t.} \quad & \frac{1}{1+\delta}S_h^* \geq e^{-r\tau}(R_h(L^*) - W_l^*) \end{aligned} \quad (\text{IC}^2)$$

where (IC<sup>2</sup>) is the incentive constraint for the high-type. Note that (IC<sup>2</sup>) is binding at the optimum; we therefore obtain  $\tau^*$  directly from (IC<sup>2</sup>):

$$\tau^* = \left( \ln \frac{(1+\delta)(R_h(L^*) - W_l^*)}{S_h^*} \right) / r > 0$$

When the low-type firm signals its type by delay,  $\tau^*$  should be long enough so that the high type will not mimic. With the offer  $(W_l^*, L_l^*, \tau^*)$ , the participation constraint for the labour union is satisfied automatically. Let  $\lambda \equiv e^{-r\tau^*}$ , and we obtain  $\pi_l(W_l^*, L_l^*, \tau^*) = \frac{1}{1+\delta}\lambda S_l^*$ . We thus obtain the following Proposition:

*Proposition 2. In the negotiation game, when the low-type firm signals by delay, there is a unique least-cost separating equilibrium: (1) the high-type firm always offers its full-information contract  $(W_h^*, L_h^*, 0)$ ; (2) the low-type firm offers the contract  $(W_l^*, L_l^*, \tau^*)$ ; (3) the union's beliefs are  $\rho(W_h^*, L_h^*, 0) = 1, \rho(W_l^*, L_l^*, \tau^*) = 0$ ; (4) the out-of-equilibrium belief of the union is  $\rho'' = 1$ .*

Suppose now the low-type firm can choose between the two signals. It will compare the payoffs with different signals to decide which signal to choose. Hence, we obtain the following Proposition:

*Proposition 3.* In the negotiation game, if  $\frac{R_l(L_l^*) - C(L_l^*)}{R_l(L_l^*) - C(L_l^*)} > \frac{\lambda + \delta}{1 + \delta}$ , the low-type firm will signal by the employment level; if  $\frac{R_l(L_l^*) - C(L_l^*)}{R_l(L_l^*) - C(L_l^*)} < \frac{\lambda + \delta}{1 + \delta}$ , it will signal by delay; if  $\frac{R_l(L_l^*) - C(L_l^*)}{R_l(L_l^*) - C(L_l^*)} = \frac{\lambda + \delta}{1 + \delta}$ , it is indifferent between the two signals.

In other words, the low-type firm will choose the signal which gives it higher payoff while separating itself from the high-type. When the low-type signals by the employment level distortion, it actually bears most signalling costs. When the low-type firm signals by delay, it shares the signalling costs with the union, but delay is more costly. Under certain conditions, the low-type prefers one way over the other.

## II.2 Examples

In the following, we discuss two examples on the firm–union negotiation with private information to illustrate the above results. Each of the examples has specific functional forms.

1. Suppose  $R_i(L) = \theta_i L$ , where  $i \in \{l, h\}$ , and  $C(L)$  is quadratic with  $C(0) = 0$ . We show that the low-type will signal only through the employment level in this case.

Suppose the low-type firm proposes the contract  $(W_l, L_l, \tau)$ , it faces the following program:

$$\begin{aligned} \max_{\{W_l, L_l, \tau\}} \quad & e^{-r\tau} (\theta_l L_l - W_l) \\ \text{s.t.} \quad & W_l - C(L_l) \geq \frac{\delta}{1 + \delta} S_l^*, \\ & \frac{1}{1 + \delta} S_h^* \geq e^{-r\tau} (\theta_h L_l - W_l) \end{aligned}$$

The second constraint is binding. Replacing  $e^{-r\tau}$  and dropping the constant terms in the objective function, the above program can be rewritten as:

$$\begin{aligned} \max_{\{W_l, L_l\}} \quad & 1 - \frac{(\theta_h - \theta_l)L_l}{\theta_h L_l - W_l} \\ \text{s.t.} \quad & W_l - C(L_l) \geq \frac{\delta}{1 + \delta} S_l^* \\ & \theta_h L_l - W_l \geq \frac{1}{1 + \delta} S_h^* \end{aligned}$$

The second constraint is because of the fact that  $e^{-r\tau} \in (0, 1]$ . If it is binding at the optimum, it means that  $\tau = 0$ . Notice that the first constraint is binding, otherwise  $W_l$  could be reduced. Thus, replacing  $W_l$ , the above program can be simplified as:

$$\begin{aligned} \min_{\{L_l\}} \quad & \frac{L_l}{(1 + \delta)[\theta_h L_l - C(L_l)] - \delta S_l^*} \\ \text{s.t.} \quad & \theta_h L_l - C(L_l) \geq \frac{S_h^* + \delta S_l^*}{1 + \delta} \end{aligned}$$

Define function  $g(L) = \theta_h L_l - C(L_l)$ , and it is strictly concave. Now we can see that the solution to the above program should lie in  $[\hat{L}_l, L_h^*)$ , where  $\hat{L}_l$  is the lower root of the equation:  $g(L) = \frac{S_h^* + \delta S_l^*}{1 + \delta}$ . It is well defined since  $g(L_h^*) > \frac{S_h^* + \delta S_l^*}{1 + \delta}$  and  $g(0) = 0$ . So we only need to restrict

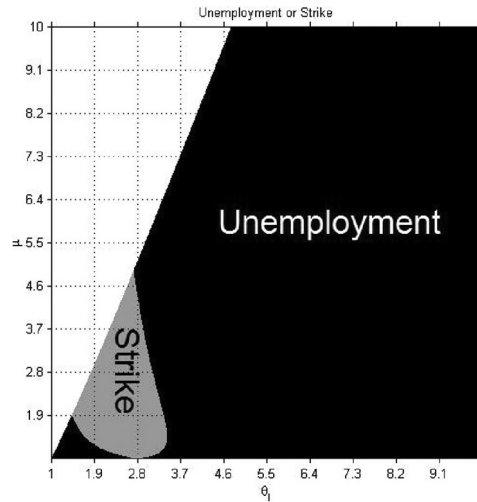


Fig. 1. Unemployment or strike.

our attention to the constraint set  $[\hat{L}_l, L_h^*)$ , and the above program can be further reduced as:

$$\begin{aligned} & \min_{\{L_l\}} \frac{C(L_l)}{L_l} \\ & s.t. \quad L_l \in [\hat{L}_l, L_h^*) \end{aligned}$$

The objective function is strictly increasing for  $L_l \in [\hat{L}_l, L_h^*)$  given that  $C(L_l)$  is quadratic with  $C(0) = 0$ . Therefore, we have a boundary solution  $L_l' = \hat{L}_l$ , and the corresponding  $\hat{W}_l = \frac{\delta}{1+\delta} S_l^* + C(\hat{L}_l)$ . Hence, we have the following result:

*Proposition 4. In the negotiation game, given that  $R_i(L) = \theta_i L$  and  $C(L)$  is quadratic with  $C(0) = 0$ , the low-type signals its type by contract  $(\hat{W}_l, \hat{L}_l, 0)$ , and thus no delay happens.*

2. Suppose  $R_i(L) = \theta_i L$  and  $C(L) = e^L$ , where  $\theta_i > 1$ , and  $i \in \{l, h\}$ .

Let  $\mu \equiv \frac{\theta_h}{\theta_l} > 1$ . Figure 1 shows the numerical results. The vertical axis represents  $\mu$ ,  $1 < \mu < 10$ , and the horizontal axis represents  $\theta_l$ ,  $0 < \theta_l < 10$ . In Figure 1, the black area shows the set of the parameter values under which the low-type firm will signal by the employment level distortion (unemployment); and the grey area identifies the set of parameter values with which the low-type will signal by delay (or enduring strikes); in the boundary of these two areas, the low-type firm is indifferent between these two signals.

The above two examples show that given specific information about the functional forms and the parameter values, the low-type firm can decide which signal to choose based on the cost–benefit criterion. Alternatively, given these conditions, one can also predict immediate settlement or strikes under our framework.

### II.3 Extension

We extend our model to the case with finite many types, and then show a numerical example, to illustrate the result.

## 1. Finite many types.

Suppose that the firm has a finite number of types:  $\theta_1 > \theta_2 > \dots > \theta_n$ , with  $n \geq 3$ , and the hazard rate is non-decreasing. Since the single-crossing property holds in our model, we apply Cho and Sobel (1990) to select the unique least-cost separating equilibrium outcome, which can be obtained by successively solving the following programs:

Program for  $\theta_1$ :

$$\begin{aligned} \max_{(W_1, L_1)} \pi_1 &= R_1(L_1) - W_1 \\ \text{s.t.} \quad W_1 - C(L_1) &\geq \frac{\delta}{1 + \delta} S_1^* \end{aligned} \quad (\text{PC}_1)$$

Program for  $\theta_k$ ,  $k = 2 \dots n$ :

$$\begin{aligned} \max_{\{(W_k, L_k), \tau_k\}} \pi_k &= \delta^{\tau_k} [R_k(L_k) - W_k] \\ \text{s.t.} \quad W_k - C(L_k) &\geq \frac{\delta}{1 + \delta} S_k^* \end{aligned} \quad (\text{PC}_k)$$

$$R_{k-1}(L_{k-1}) - W_{k-1} \geq \delta^{\tau_k} (R_k(L_k) - W_k) \quad (\text{IC}_{k-1})$$

We only need to consider the local downward incentive constraints due to the single-crossing property. As we have discussed above, different types can signal either by employment level or by delay. In particular, if those lowest types which cannot afford to distort their employment levels downward enough will signal their types only by strategic delays.

Solving the problems, we can find each type firm's offer  $m_i(W_i, L_i, \tau_i)$  and its corresponding payoff under this contract  $\pi_i(m_i)$ ,  $i = 1, \dots, n$ . We can see that only type  $\theta_1$  offers the efficient contract, that is, the first-best employment level without delay; other types offer the inefficient contracts involving either employment level distortion or delay. Furthermore, we have the following result:

*Proposition 5. There exists a threshold type  $\theta_r$  such that if  $\theta_i > \theta_r$ , type  $\theta_i$  firm will signal its type by employment level without delay; if  $\theta_i < \theta_r$ , type  $\theta_i$  firm will signal its type involving delay, while the threshold type  $\theta_r$  is indifferent between the two signals.*

Note that for all  $\theta_i > \theta_r$ , their contracts are ordering in terms of employment level, that is, higher type offers higher employment level and only the highest type offers the efficient employment level. For all  $\theta_i < \theta_r$ , their contracts are ordering in terms of delay. Those lowest types which cannot afford to distort their employment level downward will signal their types only by strategic delay. We will show a numerical example below to illustrate these results.<sup>4</sup>

## 2. A numerical example.

Assume  $n = 901$ ,  $\theta_1 = 10$ ,  $\theta_{901} = 1$ ,  $\theta_{k-1} - \theta_k = 0.01$ ,  $R_i(L_i) = \theta_i L_i$ , and  $C(L_i) = \frac{L_i^2}{2}$ , where  $k = 2, 3, \dots, 901$ ;  $i = 1, 2, \dots, 901$ ; and  $\delta = 0.95$ .

<sup>4</sup> For the continuous-type case, that is,  $\theta \in [\underline{\theta}, \bar{\theta}]$ , similar to Proposition 5, we can find a threshold type  $\hat{\theta}$  such that when  $\theta \in [\hat{\theta}, \bar{\theta}]$ , type  $\theta$  firm will signal its type by the employment level without delay; when  $\theta \in [\underline{\theta}, \hat{\theta}]$ ,  $\theta$  will separate itself by contract involving delay.

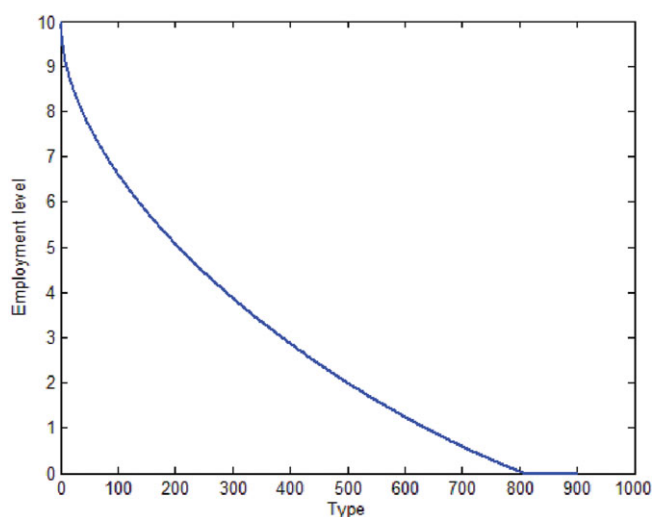


Fig. 2. Signalling by employment level.

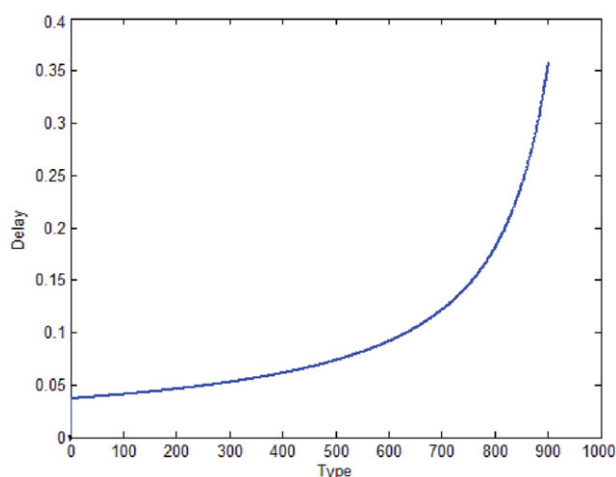


Fig. 3. Signalling by delay.

Figure 2 shows the different employment levels for different types when the employment level distortion is the only signalling device. The lower the type, the lower is the employment level. We find that for those lowest types, their employment levels are zeros, which simply means that they cannot signal their types by the employment level. Rather, they would like to use delay as the signaling device.<sup>5</sup>

Figure 3 shows the situation when delay is the only signalling device. Clearly, the lower the type, the longer the delay is.

Figure 4 shows the payoffs for different types when they signal either by the employment level or by delay. The intersection of the two payoff curves gives the threshold type. In the figure, (X: 528; Y: 1.329) means that the threshold type is type no. 528, and its corresponding payoff is

<sup>5</sup> If delay is not an option, these firms might exit the market.

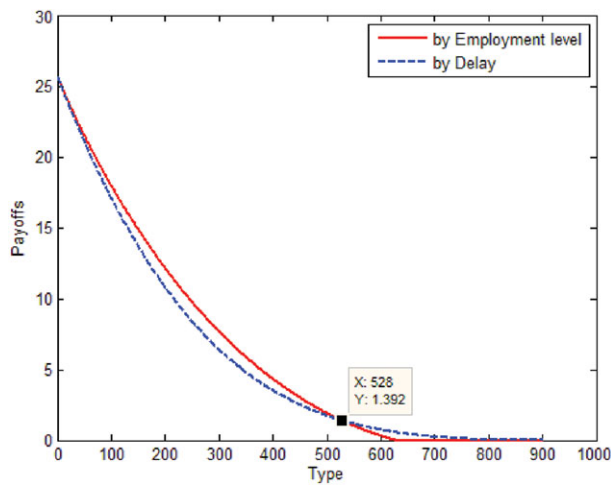


Fig. 4. Payoffs for the types when signalling by either employment level of delay.

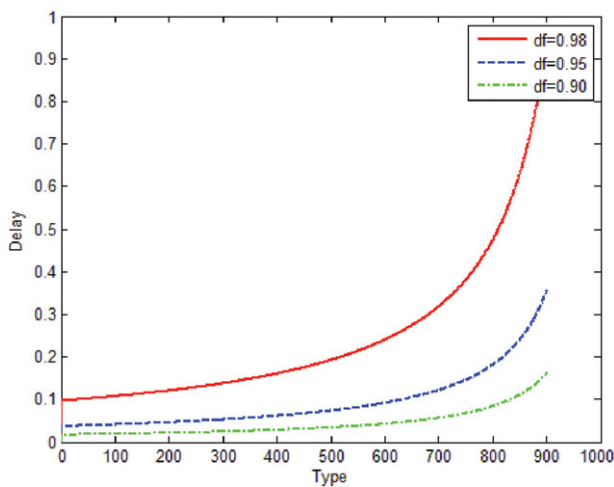


Fig. 5. Signalling by delay with different discount factors.

1.329. Above the threshold type, the firm prefers to signal by the employment level since payoff is higher; below the threshold type the firm will signal by delay, as Proposition 5 predicts.

In Figure 5, we compare delay with different discount factors, that is,  $\delta = 0.98, 0.95, 0.90$ . Intuitively, the higher the discount factor, the longer is the delay that is needed to signal the types.

In Figure 6, we attempt to identify the different threshold types with different discount factors. In Figure 6a, the discount factor is 0.98; in Figure 6b, the discount factor is 0.95; in Figure 6c, the discount factor is 0.90. It is clear that the larger the discount factor, the higher is the threshold type. Putting it a different way, as the discount factor becomes larger, more and more lowest types prefer to signal by delay as delay becomes less costly.

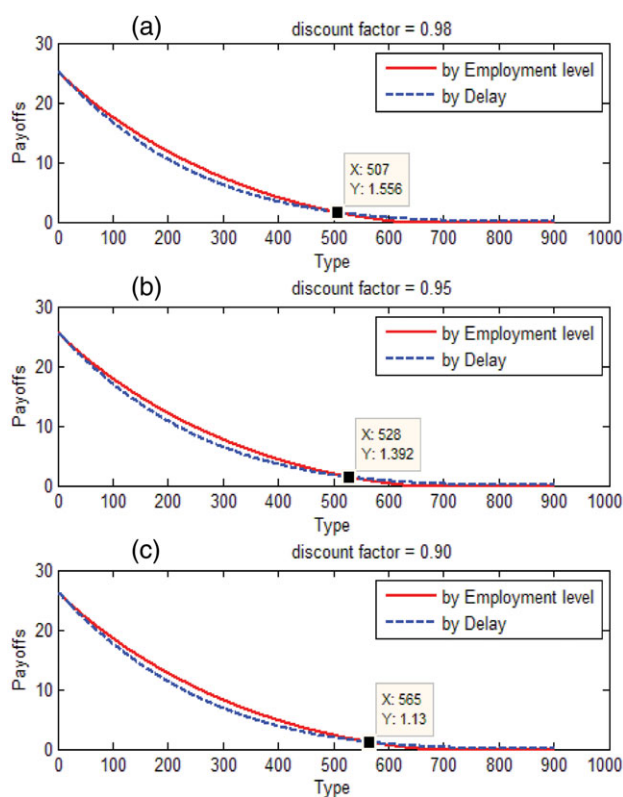


Fig. 6. Discount factors.

### III. CONCLUSION

Empirical evidence indicates that some labour contract negotiations involve strikes while most of them conclude with immediate settlement. Therefore, one needs models capable of predicting both strikes and immediate settlement, with the latter in the majority of cases. This article offers a model of union–firm negotiation with private information to show that both strikes and immediate settlement are possible in the equilibrium. The informed firm can signal its type either by the employment level or by delay. One way may dominate the other under certain conditions. When the firm uses the employment level as the signal, the result is an immediate settlement; when the firm delays its offer to signal its type, the outcome is a strike or enduring a strike.

In most signalling literature, the signalling devices are exogenously given. However, this article endogenizes the signals. We compare two signalling devices, the employment level and the strategic delay, in a single framework, the union–firm negotiation. We show that the low-type firm will choose the signal based on a cost–benefit criterion. That is, it will choose the signal which gives it higher payoff while separating itself from the high-type firm.

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